Neural Networks in Four Small Steps

A one-page explainer for store managers and other non-technical teammates. By Kirti Mittal, November 2026.

1 The goal

Teach a computer to look at a product photo and say "jacket" or "not a jacket." Nobody writes rules like "puffy plus long sleeves equals jacket." The network learns them from labeled examples.

2 A neuron is a tiny scorer

Each neuron looks at a few clues and weighs how much each one matters.

Clues: puffy = 1, long sleeves = 1, shiny = 0 | Weights: 0.8, 0.5, 0.1
Score = (1 × 0.8) + (1 × 0.5) + (0 × 0.1) = **1.3** → over 1, so "probably a jacket"

3 Neurons work in layers

Early layers spot simple things, and later layers combine them into bigger ideas.

Photo pixels → Layer 1: edges, colors → Layer 2: sleeves, zippers → Output: jacket?

4 It learns from mistakes

The network guesses, compares its answer with the correct one, and nudges every weight a little in the direction that would have been less wrong. After thousands of photos, the weights get good at the job. In most real products this happens during training, not live, so a mistake usually stops only after the team retrains the model.

Quick check: The network keeps calling raincoats "jackets." What is it learning from, and how would you fix it?
Answer: it learns from its training examples, so add more labeled raincoat photos and retrain.

What this simple version leaves out: real networks have thousands to millions of weights, and neurons do not look at named clues like "puffy." They find their own patterns, which is why their decisions can be hard to explain.

Adapted from my AIML-500 Workshop Three discussion on communication for learning. The first draft was generated with an AI assistant as the assignment required; I chose the audience and example, checked the accuracy, and edited it. Reference: IBM. (n.d.). What is a neural network? https://www.ibm.com/think/topics/neural-networks